

# An Analysis of CLIP neural network

CLIP neural network

$$a^{(l)} = \sigma(W^{(l)}a^{(l-1)} + b^{(l)})$$

## CLIP neural network -- Part 1

**Benchmarks** Classification capabilities of spiking networks trained according to unsupervised learning methods have been tested on benchmark datasets such as Iris, Wisconsin Breast Cancer or Statlog Landsat dataset. Various approaches to information encoding and network design have been used such as a 2-layer feedforward network for data clustering and classification. Based on Hopfield (1995) the authors implemented models of local receptive fields combining the properties of radial basis functions and spiking neurons to convert input signals having a floating-point representation into a spiking representation.

## CLIP neural network -- Part 2

**Temporal coding;** generating one spike per neuron, in which spike latency is inversely proportional to the pixel intensity. **Rate coding;** converting pixel intensity into a spike train, where the number of spikes is proportional to the pixel intensity. **Direct coding;** using a trainable layer to generate a floating-point value for each time step. The layer converts each pixel at a certain time step into a floating-point value, and then a threshold is used on the generated floating-point values to pick either zero or one. **Phase coding;** encoding temporal information into spike patterns based on a global oscillator. **Burst coding;** transmitting spikes in bursts, increasing communication reliability.

## CLIP neural network -- Part 3

**Hardware Efforts** to implement hardware-based spiking neural networks (SNNs) began in the 1980s, when researchers began exploring brain-inspired neuromorphic systems. In the following decades, advancements in semiconductor technologies enabled the development of several notable projects. One such project is SpiNNaker, developed at the University of Manchester, which utilizes millions of processing cores for large-scale simulation of spiking neurons. TrueNorth, developed by IBM, is one of the first commercial neuromorphic chips, designed for energy-efficient and parallel processing. Loihi, an Intel research chip, focuses on online learning and adaptability in neuromorphic modeling. In research contexts, platforms such as BrainScaleS, developed in Europe, integrate analog and digital circuitry to accelerate neural simulations. Neurogrid, from Stanford University, was designed to efficiently simulate biological neurons and synapses. Dynap-se models developed by iniLabs,

are a family of low-power, event-based neuromorphic chip intended for use in robotics and Internet of Things (IoT) applications. In addition, hardware based on memristors and other emerging memory technologies is being explored for the implementation of SNNs, with the goal of achieving lower power consumption and improved compatibility with biological neural models.

## CLIP neural network -- Part 4

SNNs can in principle be applied to the same applications as traditional ANNs. In addition, SNNs can model the central nervous system of biological organisms, such as an insect seeking food without prior knowledge of the environment. Due to their relative realism, they can be used to study biological neural circuits. Starting with a hypothesis about the topology of a biological neuronal circuit and its function, recordings of this circuit can be compared to the output of a corresponding SNN, evaluating the plausibility of the hypothesis. SNNs lack effective training mechanisms, which can complicate some applications, including computer vision. When using SNNs for image based data, the images need to be converted into binary spike trains. Types of encodings include:

## CLIP neural network -- Part 5

Spiking neural networks (SNNs) are artificial neural networks (ANN) that mimic natural neural networks. These models leverage timing of discrete spikes as the main information carrier. In addition to neuronal and synaptic state, SNNs incorporate the concept of time into their operating model. The idea is that neurons in the SNN do not transmit information at each propagation cycle (as it happens with typical multi-layer perceptron networks), but rather transmit information only when a membrane potential—an intrinsic quality of the neuron related to its membrane electrical charge—reaches a specific value, called the threshold. When the membrane potential reaches the threshold, the neuron fires, and generates a signal that travels to other neurons which, in turn, increase or decrease their potentials in response to this signal. A neuron model that fires at the moment of threshold crossing is also called a spiking neuron model. While spike rates can be considered the analogue of the variable output of a traditional ANN, neurobiology research indicated that high speed processing cannot be performed solely through a rate-based scheme. For example humans can perform an image recognition task requiring no more than 10ms of processing time per neuron through the successive layers (going from the retina to the temporal lobe). This time window is too short for rate-based encoding. The precise spike timings in a small set of spiking neurons also has a higher information coding

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capacity compared with a rate-based approach. The most prominent spiking neuron model is the leaky integrate-and-fire model. In that model, the momentary activation level (modeled as a differential equation) is normally considered to be the neuron's state, with incoming spikes pushing this value higher or lower, until the state eventually either decays or—if the firing threshold is reached—the neuron fires. After firing, the state variable is reset to a lower value. Various decoding methods exist for interpreting the outgoing spike train as a real-value number, relying on either the frequency of spikes (rate-code), the time-to-first-spike after stimulation, or the interval between spikes.

## CLIP neural network -- Part 6

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resorting to entirely biologically inspired local learning rules for the hidden units translating conventionally trained "rate-based" NNs to SNNs smoothing the network model to be continuously differentiable defining an SG (Surrogate Gradient) as a continuous relaxation of the real gradients The second concerns the optimization algorithm. Standard BP can be expensive in terms of computation, memory, and communication and may be poorly suited to the hardware that implements it (e.g., a computer, brain, or neuromorphic device). Incorporating additional neuron dynamics such as Spike Frequency Adaptation (SFA) is a notable advance, enhancing efficiency and computational power. These neurons sit between biological complexity and computational complexity. Originating from biological insights, SFA offers significant computational benefits by reducing power usage, especially in cases of repetitive or intense stimuli. This adaptation improves signal/noise clarity and introduces an elementary short-term memory at the neuron level, which in turn, improves accuracy and efficiency. This was mostly achieved using compartmental neuron models. The simpler versions are of neuron models with adaptive thresholds, are an indirect way of achieving SFA. It equips SNNs with improved learning capabilities, even with constrained synaptic plasticity, and elevates computational efficiency. This feature lessens the demand on network layers by decreasing the need for spike processing, thus lowering computational load and memory access time—essential aspects of neural computation. Moreover, SNNs utilizing neurons capable of SFA achieve levels of accuracy that rival those of conventional ANNs, while also requiring fewer neurons for comparable tasks. This efficiency streamlines the computational workflow and conserves space and energy, while maintaining technical integrity. High-performance deep spiking neural networks can operate with 0.3 spikes per neuron.